

Notes: French and Roll (JFE, 1986)

Stock Return Variances: The Arrival of Information and the Reaction of Traders

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Sample	3
3.2	Variance-ratio construction	3
3.3	Trading and non-trading intervals	3
3.4	Autocorrelations and long-horizon returns	4
4	Models and empirical specifications	4
4.1	Trading versus non-trading hourly variance	4
4.2	Three hypotheses	4
4.3	Daily return decomposition	4
4.4	Bid-ask error mechanics	5
4.5	Long-horizon variance-ratio test	5
5	Identification and empirical design	5
5.1	What identifies the role of trading	5
5.2	Public versus private information	5
5.3	Private information versus trading noise	6
5.4	Limitations	6
6	Section-by-section study guide	6
7	Tables and theoretical result blocks	7
7.1	Tables 1 and 2: Non-trading variance and exchange-holiday variance	7
7.2	Tables 3 and 4: Daily autocorrelations and actual-to-implied variance ratios	7
7.3	Table A.1: Exchange holiday information consistent with information variance exceeding mispricing variance	8
8	Theoretical result blocks	9
8.1	Result Block 1: Trading time versus calendar time	9
8.2	Result Block 2: Exchange holidays identify trading dependence	9
8.3	Result Block 3: Mispricing is present but bounded	9
9	Detailed result map	10

10 Short summary	10
11 Conclusion	10
11.1 Core conclusion	10
11.2 Economic intuition	10
11.3 Incremental contribution revisited	11
11.4 Implications for market microstructure	11
11.5 Limitations	11
11.6 Takeaway	11

1 Big picture

- **Question.** Why are stock returns much more volatile when exchanges are open than when exchanges are closed?
- **Puzzle.** If information arrived uniformly over calendar time and prices simply incorporated that information, then longer non-trading intervals should generate proportionally larger return variances. Instead, weekend and holiday returns have surprisingly low variance relative to normal weekday returns.
- **Three candidate explanations.** The paper compares: (i) public information arrives mainly during business hours; (ii) private information affects prices mainly when informed investors can trade; and (iii) the trading process itself creates temporary pricing errors or bid-ask-related noise.
- **Core empirical strategy.** French and Roll use NYSE and AMEX daily returns from 1963–1982, weekends and holidays, the special 1968 Wednesday exchange closures, daily autocorrelations, and long-horizon actual-to-implied variance ratios.
- **Main result.** Trading-hour variance is enormously larger than non-trading variance. Weekend variance is only about 10.7 percent higher than one trading-day variance, even though the calendar interval is three days.
- **Mispricing result.** Trading noise and bid-ask effects explain a nontrivial fraction of daily variance, roughly 4 to 12 percent on average, but this is too small to explain the huge trading/non-trading variance differential.
- **Private-information result.** Exchange holidays show that when the market is closed on a business day, return variance is low. This favors the view that much volatility comes from private information being incorporated through trading.
- **Economic takeaway.** Prices move much more during trading hours because information - especially private information - is produced, revealed, and incorporated through trading. Market microstructure noise matters, but it is not the dominant source of the trading-time volatility premium.

2 Incremental contribution of the paper

- **Separates calendar time from trading time.** Earlier work documented weekend and intraday volatility patterns. French and Roll build a direct test of whether variance depends on the mere passage of time or on the market being open.
- **Distinguishes public information, private information, and trading noise.** The paper does not simply document high trading-hour volatility. It decomposes the possible explanations into three mechanisms with different testable implications.
- **Uses exchange holidays as a quasi-experiment.** The 1968 Wednesday exchange closures and election-day closures create business days when information-generating activity could continue but exchange trading could not. This is a powerful way to separate public-information arrival from trading-dependent information incorporation.
- **Introduces long-horizon variance-ratio logic for mispricing.** Comparing actual multi-day or multi-month variances with variances implied by daily returns allows the authors to estimate how much daily volatility is temporary pricing error or bid-ask bounce.
- **Quantifies microstructure noise without making it the whole story.** The paper acknowledges trading noise and estimates bounds on its contribution, but shows it has only a trivial effect on the trading versus non-trading variance gap.
- **Connects market microstructure to asset-price volatility.** The paper bridges informational market microstructure and variance-ratio asset-pricing evidence: trades reveal private information, so volatility is concentrated when trading can occur.

- **Provides a foundation for later high-frequency market-quality research.** Later studies on price discovery, trading halts, market closures, after-hours announcements, and information production build on this basic distinction between information arrival and information incorporation through trading.
- **Refines interpretations of return autocorrelation.** The autocorrelation evidence links bid-ask bounce, delayed adjustment, and pricing errors to observed daily return dynamics, especially across firm-size quintiles.

3 Data and measurement

3.1 Sample

- The data are daily returns from the Center for Research in Security Prices for all common stocks listed on the New York and American Stock Exchanges.
- The sample period runs from January 1963 through December 1982.
- The paper divides the twenty-year sample into ten two-year subperiods.
- Stocks are sorted into equity-value quintiles at the beginning of each subperiod.

Interpretation: The sample is broad enough to examine market-wide variance patterns and size-related heterogeneity. The two-year subperiods reduce the influence of long-run changes in volatility and market structure.

3.2 Variance-ratio construction

- Table 1 compares multiple-calendar-day returns, such as weekend and holiday returns, with single-day trading returns.
- Table 2 compares exchange-holiday returns with normal one-day and normal weekly returns.
- Table 4 compares actual long-horizon return variances with variances implied by daily returns under independence.
- The logic is: if volatility accumulates linearly with calendar time, multi-day variances should scale with time. If volatility is concentrated in trading time, multi-day non-trading intervals will have low ratios.

Interpretation: Variance ratios are the paper’s main measurement tool. They translate the qualitative idea of “trading-time volatility” into numerical comparisons across holding periods.

3.3 Trading and non-trading intervals

- A normal weekday return includes a trading period and an overnight non-trading period.
- A weekend return includes a long non-trading period plus the Monday trading period.
- Holiday and holiday-weekend returns extend the non-trading period even further.
- Exchange holidays in 1968 are especially informative because they were business days when the exchanges were closed.

Interpretation: Weekends confound exchange closure with weekend business inactivity. The 1968 Wednesday exchange holidays help separate exchange closure from nonbusiness time.

3.4 Autocorrelations and long-horizon returns

- Daily autocorrelations are estimated by stock and subperiod, then averaged across stocks and subperiods.
- Positive daily autocorrelations among large firms suggest temporary delayed incorporation or overreaction patterns that are not simply bid-ask bounce.
- Long-horizon actual-to-implied variance ratios reveal whether daily volatility contains temporary components that disappear over longer horizons.

Interpretation: Autocorrelation and long-horizon variance evidence are used to measure the importance of trading noise relative to information.

4 Models and empirical specifications

4.1 Trading versus non-trading hourly variance

Let σ_T^2 denote hourly variance during exchange trading hours and σ_N^2 hourly variance during non-trading hours. For a weekend return, the paper uses the relation

$$66\sigma_N^2 + 6\sigma_T^2 = 1.107(18\sigma_N^2 + 6\sigma_T^2),$$

which implies a trading-to-non-trading hourly variance ratio of roughly 71.8 for the average weekend comparison.

Interpretation: This calculation turns the low weekend-to-weekday variance ratio into a striking hourly variance comparison. Most volatility is concentrated in the short trading window rather than in the long weekend interval.

4.2 Three hypotheses

1. **Public information hypothesis:** high trading-time volatility is caused by public information arriving mostly during business hours.
2. **Private information hypothesis:** high trading-time volatility is caused by private information being incorporated when informed investors trade.
3. **Trading noise hypothesis:** high trading-time volatility is caused by pricing errors, bid-ask effects, or other noise generated by trading.

Interpretation: The hypotheses are not mutually exclusive. The paper's objective is to evaluate their relative empirical importance.

4.3 Daily return decomposition

The paper later writes daily returns as the sum of an information component, a mispricing component, and a bid-ask error component:

$$R_t = X_t + Y_t + Z_t.$$

Here X_t is the rational information component, Y_t is mispricing, and Z_t is bid-ask error.

Interpretation: The decomposition is useful because X_t should persist in long-horizon returns, while Y_t and Z_t should largely disappear as horizons lengthen.

4.4 Bid-ask error mechanics

The bid-ask error in daily return can be written as

$$Z_t = e_t - e_{t-1}.$$

If daily price errors e_t are independent, then

$$\text{var}(Z_t) = 2 \text{var}(e_t), \quad \text{cov}(Z_t, Z_{t-1}) = -\text{var}(e_t).$$

Interpretation: Bid-ask bounce mechanically produces negative first-order autocorrelation. This helps interpret why small firms, where bid-ask effects are more important, can show negative short-lag autocorrelations.

4.5 Long-horizon variance-ratio test

Let v_L be an actual-to-implied variance ratio for a long horizon, such as six months. If pricing errors and bid-ask errors are temporary, then a long-horizon return mostly reflects information. A low actual-to-implied ratio means daily variance contains a temporary component.

Interpretation: Table 4 uses this logic to estimate the fraction of daily variance that disappears at longer horizons.

5 Identification and empirical design

5.1 What identifies the role of trading

- Ordinary weekends show that the mere passage of calendar time does not generate proportional variance.
- Exchange holidays show that volatility falls when the market is closed even on business days.
- Weekly ratios around holidays show whether volatility merely shifts to following trading days.
- Long-horizon variance ratios show whether daily variance is dominated by temporary noise.

Interpretation: The empirical design is not a single regression. It is a sequence of tests that progressively narrows the explanation for high trading-time variance.

5.2 Public versus private information

- If public information arrives during business hours regardless of exchange trading, Wednesday exchange holidays should have high variance.
- The paper finds low two-day exchange-holiday variance relative to normal business-day variance.
- This weakens the pure public-information explanation.

Interpretation: Public information is not ruled out completely, but it cannot explain why exchange closure on business days sharply lowers return variance.

5.3 Private information versus trading noise

- Both private-information and trading-noise hypotheses predict low variance when the exchange is closed.
- Long-horizon variance ratios estimate how much daily volatility is temporary trading noise.
- The estimated noise component is too small to explain the enormous trading-time volatility premium.

Interpretation: This is the key identifying distinction. Trading noise exists, but not enough to be the dominant explanation.

5.4 Limitations

- The paper uses daily close-to-close returns, not intraday quote and trade data.
- The 1968 exchange holidays were unusual and related to operational backlogs.
- The model is intentionally simple and decomposes returns into stylized components.
- The paper cannot observe private information directly; it infers its role from variance patterns.

Interpretation: The inference is indirect but powerful because the competing hypotheses imply different variance behavior around closures and at long horizons.

6 Section-by-section study guide

- **Introduction.** Establishes the trading-time volatility puzzle and previews the three mechanisms.
- **Trading and non-trading variances.** Uses weekends and holidays to show that return variance does not scale with calendar time.
- **Possible explanations.** Defines public information, private information, and trading noise hypotheses.
- **Exchange holidays.** Uses 1968 Wednesday closures and election holidays to test whether volatility falls when the exchange is closed.
- **Daily autocorrelations.** Uses autocorrelation patterns to identify bid-ask bounce and possible delayed adjustment or overreaction.
- **Long-horizon variances.** Estimates how much daily variance disappears over longer holding periods.
- **Appendix.** Explores correlations between information and mispricing components and bounds on information production during exchange holidays.

7 Tables and theoretical result blocks

7.1 Tables 1 and 2: Non-trading variance and exchange-holiday variance

Read:

- Table 1 shows weekend and holiday return variances are only slightly above normal one-day variances.
- The weekend average variance ratio is 1.107, far below the value of 3 implied by calendar-time variance scaling.
- Table 2 shows that two-day exchange-holiday returns in 1968 have an average daily variance ratio of 1.145.
- Weekly ratios around exchange holidays are below one, showing that the variance lost during closure is not fully recovered within the week.

Detailed interpretation: Table 1 establishes the central puzzle: stock variance is concentrated in trading periods. Table 2 then sharpens identification by using exchange holidays on business days. If public information arrived normally on those Wednesdays, variance should not have fallen so much. The low variance instead points toward information being incorporated through trading.

Economic message: Trading hours are not just windows in which already-arrived public information is recorded. They are when private information is produced, revealed, and impounded into prices.

Table 1

Average ratios of multiple-day variances relative to single-day variances for all NYSE and AMEX stocks and for quintiles of stocks sorted by equity value.^a

		All stocks	Smallest quintile ^a	2	3	4	Largest quintile
Two-day holidays	Average ratio ^b	1.247	1.301	1.199	1.239	1.217	1.281
	Standard error ^c	0.066	0.068	0.054	0.052	0.097	0.100
	Standard deviation ^d	1.354	1.446	1.270	1.371	1.149	1.351
	Number of firms ^e	1962.5	390.3	392.3	392.4	393.5	394.1
	Average sample size ^f	10.0	10.0	10.0	10.0	10.0	10.0
Weekends	Average ratio	1.107	1.122	1.108	1.119	1.105	1.082
	Standard error	0.012	0.010	0.016	0.014	0.014	0.017
	Standard deviation	0.385	0.412	0.379	0.435	0.337	0.286
	Number of firms	2055.3	411.2	410.8	410.6	411.2	411.5
	Average sample size	92.8	92.5	92.8	92.9	93.0	93.0
Holiday weekends	Average ratio	1.117	1.111	1.122	1.099	1.122	1.130
	Standard error	0.092	0.053	0.085	0.071	0.106	0.151
	Standard deviation	1.219	1.176	0.992	1.276	1.232	1.014
	Number of firms	2055.7	411.3	410.8	410.9	411.2	411.5
	Average sample size	11.1	11.0	11.1	11.1	11.1	11.1

^a Stocks are sorted into quintiles based on their equity values at the beginning of ten two-year subperiods between 1963–1982.

^b The average ratio comparing the variance of two-, three-, and four-calendar-day returns with the variance of one-day returns. This estimate is the average of the ten subperiod averages.

^c The standard error of the reported average ratio. This standard error is based on the distribution of the ten subperiod average ratios.

^d The average cross-sectional standard deviation. The ratios for individual firms are used to estimate the standard deviation for each subperiod. The reported standard deviation is the average of the ten subperiod standard deviations.

^e The average number of firms in each subperiod.

^f The average number of multiple-day returns for each stock in each subperiod.

(a) Table 1: Average ratios of multiple-day variances relative to single-day variances

Table 2

Daily and weekly variance ratios for exchange holidays.

		All stocks	Smallest quintile ^a	2	3	4	Largest quintile
<i>Daily variance ratios</i>							
Exchange holidays in 1968	Average ratio ^b	1.145	1.077	1.043	1.180	1.239	1.274
	Standard deviation ^c	0.882	0.857	0.647	0.979	0.944	1.001
	Number of firms	2083	597	455	374	342	315
	Average sample size ^d	22.7	22.8	22.7	22.6	22.7	22.85
Election holidays	Average ratio ^e	1.165	1.131	1.073	1.186	1.159	1.332
	Standard deviation	1.079	1.222	1.065	1.040	0.799	1.118
	Number of firms	2026	572	426	367	347	314
	Average sample size	8.5	8.2	8.2	8.1	8.7	9.5
<i>Weekly variance ratios</i>							
Exchange holidays in 1968	Average ratio ^f	0.821	0.901	0.802	0.772	0.793	0.784
	Standard deviation	0.559	0.667	0.484	0.422	0.511	0.612
	Number of firms	2093	600	457	376	344	316
	Average sample size	20.6	20.7	20.6	20.5	20.7	20.8
Election holidays	Average ratio	0.839	0.876	0.776	0.889	0.779	0.868
	Standard deviation	0.614	0.707	0.627	0.678	0.501	0.527
	Number of firms	1188	278	221	192	229	268
	Average sample size	8.5	8.3	8.3	8.3	8.6	8.8

^a Firms are sorted into quintiles based on their relative equity values when they are first listed in the CRSP daily master file.

^b Average variance ratio comparing two-day exchange holiday returns with single-calendar-day returns between January 1963 and December 1982.

^c Cross-sectional standard deviation of the individual firm ratios.

^d Average number of exchange holidays for each firm.

^e Average variance ratio comparing two-day exchange holiday returns with single-calendar-day returns from 1962–1969, 1972, 1976, and 1980.

(b) Table 2: Daily and weekly variance ratios for exchange holidays

7.2 Tables 3 and 4: Daily autocorrelations and actual-to-implied variance ratios

Read:

- Table 3 reports average daily autocorrelations at lags 1–15.
- Small firms show negative first-order autocorrelation, consistent with bid-ask bounce.
- Large firms show positive first-order autocorrelation, consistent with delayed adjustment or persistent pricing errors.

- Table 4 reports actual-to-implied variance ratios from two days through six months.
- The six-month ratio is 0.883 for all stocks, 0.731 for the smallest quintile, and 0.907 for the largest quintile.

Detailed interpretation: Table 3 confirms that daily returns contain short-run dynamics, but the magnitudes are not large enough to explain the main variance puzzle. Table 4 quantifies the temporary component. If daily variance were entirely rational information, actual long-horizon variance should equal implied variance. Ratios below one mean some daily variance is temporary. The temporary component is larger for small firms but still not enough to explain the trading-hour volatility premium.

Economic message: Trading noise is real, especially for small firms, but it is a secondary explanation. Information flow, especially private information revealed through trading, remains the main driver.

Table 3
Average daily autocorrelations in percent for all NYSE and AMEX stocks and for quintiles of stocks sorted by equity value.^a

Lag	All stocks	Smallest quintile	2	3	4	Largest quintile
1	0.33 (0.87)	-6.42 (1.55)	-1.66 (1.07)	1.17 (0.83)	2.49 (0.84)	5.44 (1.01)
2	-1.15 (0.15)	-1.94 (0.24)	-1.43 (0.20)	-1.28 (0.15)	-0.75 (0.14)	-0.40 (0.26)
3	-1.15 (0.15)	-1.35 (0.16)	-1.39 (0.23)	-1.26 (0.19)	-1.00 (0.21)	-0.81 (0.22)
4	-0.68 (0.23)	-0.85 (0.16)	-0.66 (0.19)	-0.62 (0.29)	-0.49 (0.28)	-0.78 (0.34)
5	-0.28 (0.23)	-0.44 (0.17)	-0.15 (0.28)	-0.05 (0.22)	-0.15 (0.27)	-0.59 (0.39)
6	-0.95 (0.24)	-0.84 (0.24)	-0.72 (0.25)	-0.84 (0.29)	-0.92 (0.26)	-1.38 (0.29)
7	-0.64 (0.25)	-0.53 (0.19)	-0.53 (0.30)	-0.50 (0.27)	-0.63 (0.26)	-0.98 (0.32)
8	-0.37 (0.24)	-0.14 (0.15)	-0.42 (0.23)	-0.31 (0.30)	-0.25 (0.25)	-0.73 (0.40)
9	-0.45 (0.19)	-0.34 (0.13)	-0.47 (0.16)	-0.27 (0.17)	-0.57 (0.20)	-0.60 (0.40)
10	-0.26 (0.15)	-0.12 (0.12)	-0.10 (0.15)	-0.15 (0.18)	-0.38 (0.19)	-0.51 (0.24)
11	-0.52 (0.18)	-0.27 (0.10)	-0.54 (0.13)	-0.58 (0.21)	-0.57 (0.23)	-0.65 (0.32)
12	-0.20 (0.22)	-0.35 (0.12)	-0.18 (0.20)	-0.21 (0.24)	-0.10 (0.28)	-0.19 (0.31)
13	-0.15 (0.20)	-0.11 (0.12)	0.02 (0.18)	-0.03 (0.20)	-0.14 (0.28)	-0.47 (0.35)
14	0.15 (0.25)	0.28 (0.13)	0.05 (0.17)	0.18 (0.24)	0.14 (0.27)	0.10 (0.50)
15	0.42 (0.15)	0.28 (0.12)	0.38 (0.17)	0.49 (0.17)	0.43 (0.21)	0.52 (0.28)

^aThe autocorrelations and standard errors (in parentheses) are reported in percent. Autocorrelations are estimated for individual firms during each of ten two-year subperiods between 1963 and 1982. These autocorrelations are averaged to compute subperiod averages. Each reported autocorrelation is the average of ten subperiod averages. The standard error is based on the distribution of the ten subperiod averages. Approximately 500 returns are used to estimate the autocorrela-

(a) Table 3: Average daily autocorrelations

Table 4
Actual-to-implied variance ratios for all NYSE and AMEX stocks and for quintiles of stocks sorted by equity value.

		All stocks	Smallest quintile ^a	2	3	4	Largest quintile
Two trading days	Average ratio ^b	0.999	0.933	0.979	1.007	1.021	1.048
	Standard error ^c	0.010	0.015	0.011	0.010	0.009	0.013
	Number of firms ^d	1900.2	362.6	374.6	373.9	386.8	402.3
	Average sample size ^e	250.5	249.0	250.3	250.8	251.0	251.0
One week	Average ratio	0.966	0.853	0.928	0.979	1.005	1.053
	Standard error	0.017	0.025	0.019	0.018	0.018	0.019
	Number of firms	1899.6	362.1	374.5	373.9	386.8	402.3
	Average sample size	103.6	102.7	103.6	103.9	104.0	104.1
Two weeks	Average ratio	0.943	0.803	0.900	0.959	0.995	1.045
	Standard error	0.024	0.026	0.025	0.025	0.027	0.025
	Number of firms	1899.5	362.0	374.5	373.9	386.8	402.3
	Average sample size	51.5	50.9	51.5	51.7	51.8	51.8
Three weeks	Average ratio	0.929	0.784	0.888	0.953	0.985	1.024
	Standard error	0.022	0.026	0.026	0.023	0.024	0.024
	Number of firms	1899.5	362.0	374.4	373.9	386.8	402.2
	Average sample size	34.0	33.5	34.0	34.1	34.2	34.2
One month	Average ratio	0.906	0.773	0.874	0.931	0.959	0.983
	Standard error	0.022	0.023	0.027	0.025	0.020	0.024
	Number of firms	1898.9	361.7	374.3	373.9	386.8	402.2
	Average sample size	23.6	23.0	23.6	23.8	23.8	23.9
Three months	Average ratio	0.894	0.752	0.876	0.949	0.942	0.941
	Standard error	0.045	0.032	0.043	0.051	0.055	0.066
	Number of firms	1895.3	359.8	373.6	373.5	386.6	401.8
	Average sample size	7.7	7.3	7.7	7.8	7.9	7.9
Six months	Average ratio	0.883	0.731	0.862	0.931	0.929	0.907
	Standard error	0.102	0.062	0.086	0.109	0.117	0.129
	Number of firms	1554.2	203.4	291.8	324.8	350.5	383.7
	Average sample size	4.0	4.0	4.0	4.0	4.0	4.0

^aStocks are sorted in quintiles based on their equity value at the beginning of ten two-year subperiods between 1963–1982.

^bAverage ratio comparing the actual holding period variance with the variance implied by single-trading-day returns under the assumption that the one-day returns are independent. The reported ratio is the average of ten subperiod averages.

^cThe standard error of the reported average ratio. This standard error is based on the distribution of the ten subperiod average ratios.

(b) Table 4: Actual-to-implied variance ratios

7.3 Table A.1: Exchange holiday information consistent with information variance exceeding mispricing variance

Read:

- The appendix table maps combinations of variance ratios into bounds on information production during exchange holidays.
- For the authors' point estimates, information variance exceeds mispricing variance only over a narrow range of non-market business-day information production.
- Even with high exchange-holiday variance ratios, information production on closed business days must be well below normal-day production for information to dominate mispricing under the appendix assumptions.

Detailed interpretation: The appendix explores a more general case in which information and mispricing components may be correlated. This is a robustness and boundary exercise. It shows that the conclusion about private information does not depend simply on assuming independence, though the quantitative ranges can become narrow.

Economic message: The appendix reinforces the idea that trading is central to information production and incorporation. When the market is closed, the amount of economically relevant information impounded into prices appears much lower.

Table A.1
Exchange holiday information consistent with information variance exceeding mispricing variance
(% of normal day).

Actual-to-implied variance ratio, a	Ratio of two-day exchange holiday return variance to normal day return variance, b					
	1.05	1.10	1.15	1.20	1.25	1.30
	Information produced on exchange holiday as a percentage of information produced, on a normal day, consistent with information variance exceeding mispricing variance ^a					
0.75	5.78 to 9.88	11.59 to 19.56	17.41 to 29.07	23.26 to 38.44	29.12 to 47.68	34.99 to 56.82
0.80	5.60 to 9.86	11.21 to 19.48	16.84 to 28.90	22.48 to 38.15	28.13 to 47.26	33.80 to 56.25
0.85	5.43 to 9.84	10.87 to 19.40	16.32 to 28.73	21.78 to 37.87	27.24 to 46.86	32.72 to 55.70
0.90	5.27 to 9.82	10.55 to 19.32	15.84 to 28.87	21.13 to 37.60	26.43 to 46.46	31.73 to 55.17
0.95	5.13 to 9.79	10.27 to 19.24	15.40 to 28.40	20.54 to 37.34	25.69 to 46.08	30.83 to 54.66

^aThe lower and upper bounds are given by

Figure 3: Table A.1: Exchange holiday information consistent with information variance exceeding mispricing variance

8 Theoretical result blocks

8.1 Result Block 1: Trading time versus calendar time

- Calendar time alone predicts variance proportional to the length of the return interval.
- The data reject this strongly: weekends have much lower variance than calendar scaling implies.
- The trading window matters more than elapsed time.

Interpretation: This result is the empirical starting point for the paper. Asset-price volatility is a market-operation phenomenon, not just a passive clock-time phenomenon.

8.2 Result Block 2: Exchange holidays identify trading dependence

- Exchange holidays are business days without exchange trading.
- Public information could still arrive, but informed investors cannot trade on the exchange.
- Low exchange-holiday variance therefore points toward trading-dependent information incorporation.

Interpretation: The 1968 Wednesday closures are the paper's cleanest source of evidence against a pure public-news-arrival explanation.

8.3 Result Block 3: Mispricing is present but bounded

- Autocorrelations and long-horizon ratios imply temporary daily volatility.
- The paper estimates that roughly 4–12 percent of daily variance is caused by mispricing.
- Eliminating this effect changes the weekend-to-weekday variance ratio only slightly.

Interpretation: This result avoids a false dichotomy. Trading noise matters, but information flow matters much more.

9 Detailed result map

- **Puzzle:** trading-hour variance is much larger than non-trading-hour variance.
- **Table 1:** weekend and holiday variance ratios are far below calendar-time predictions.
- **Table 2:** exchange holidays lower variance, inconsistent with pure public-news arrival.
- **Table 3:** daily autocorrelations reveal bid-ask bounce and temporary pricing dynamics.
- **Table 4:** long-horizon variance ratios bound the amount of temporary trading noise.
- **Appendix:** information dominates mispricing only under restricted ranges of closed-day information production.
- **Conclusion:** private information incorporated through trading is the main source of the trading-time volatility premium.

10 Short summary

- French and Roll study why stock returns are much more volatile while exchanges are open.
- They compare weekday, weekend, holiday, and exchange-holiday return variances for NYSE and AMEX stocks from 1963–1982.
- Weekend variance is far lower than calendar time scaling predicts.
- Exchange holidays show that variance is low when the market is closed even on business days.
- Daily autocorrelations and long-horizon variances show that temporary pricing errors and bid-ask effects exist.
- These effects explain roughly 4–12 percent of daily variance, not the huge trading/non-trading variance gap.
- The main explanation is that private information is incorporated into prices through trading.

11 Conclusion

11.1 Core conclusion

French and Roll show that asset-price volatility is concentrated in exchange trading hours. The explanation is not simply that calendar time passes or that public news arrives during the workday. Instead, much volatility appears when informed investors can trade and private information can be incorporated into prices.

11.2 Economic intuition

Information that can be acted on is more valuable than information that cannot be traded on. When the exchange is open, analysts and investors produce private information, trade on it, and move prices. When the exchange is closed, less private information is produced and any existing private information is not incorporated into exchange prices until trading resumes.

11.3 Incremental contribution revisited

The paper’s incremental contribution is to provide a clean empirical decomposition of trading-time volatility into public information, private information, and trading noise. Its exchange-holiday design and long-horizon variance-ratio approach allow it to show that trading noise is real but not dominant, while private-information incorporation is central.

11.4 Implications for market microstructure

The findings imply that trading is not merely a mechanical matching process. Trading is part of the information-production and price-discovery process. Market closures, trading halts, and after-hours periods can therefore change the rate at which information enters prices.

11.5 Limitations

The paper uses daily returns and cannot directly observe trades, quotes, or private signals. It studies a historical period before modern electronic trading, high-frequency data, and extended-hours markets. Nevertheless, the logic remains foundational: the variance of returns depends strongly on when markets are open.

11.6 Takeaway

The central takeaway is that volatility is not simply a function of clock time. It is a function of trading institutions and information incorporation. Prices move most when informed investors can act on information.